

Name: _____

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30 minutes maximum. No aids (book, calculator, etc.) are permitted. Show all work and use proper notation for full credit. Answers should be in reasonably-simplified form. 25 points possible.

1. (10 points) Evaluate the following improper integrals, if possible. If the integral is not convergent, answer divergent. **You must use correct notation and show your work.** Answers without work will earn no credit.

(a) $\int_2^{\infty} \frac{10}{1+x^2} dx$

(b) $\int_1^5 \frac{1}{\sqrt{15-3x}} dx$

2. (2 points) **Without integrating** determine whether the integral $\int_2^{\infty} \frac{10}{10x+x^2} dx$ converges or diverges. Justify your conclusion. An answer without any work will earn no credit. (HINT: Use your work in Problem 1, above.)

3. (2 points) Find the first three terms of the sequence defined below, starting with $n = 1$.

$$a_1 = 4 \text{ and } a_n = 3a_{n-1} - 2 \text{ for } n \geq 2$$

4. (3 points) Write a general formula for the sequence $\{\frac{1}{2}, \frac{-1}{4}, \frac{1}{6}, \frac{-1}{8}, \frac{1}{10}, \frac{-1}{12}, \dots\}$. A complete answer includes **both** a formula for a_n and clear indication of where the index n starts. (Does n start at zero? one? two?)

5. (8 points) Determine the limit of the sequence or show that the sequence diverges. If it converges, find its limit. Answers with no work will earn no credit.

(a) $a_n = \frac{n}{3^n}$

(b) $a_n = \frac{\sqrt{n}}{\sqrt{2n+1}}$